

Enhanced Multi-Authority Ethical Supply Chain Tracking System Using Blockchain for Transparent Healthcare Verification

Suyash Jain

*Department of Networking and Communications
SRM Institute of Science and Technology
Kattankulathur, India
sj5889@srmist.edu.in*

Prince Bagla

*Department of Networking and Communications
SRM Institute of Science and Technology
Kattankulathur, India
pb0795@srmist.edu.in*

Godwin J Ponsam

*Department of Networking and Communications
SRM Institute of Science and Technology
Kattankulathur, India
godwinj@srmist.edu.in*

Abstract—This paper presents an enhanced blockchain-based supply chain tracking system for the healthcare sector that introduces a threshold-based multi-authority ethical validation mechanism. The proposed system addresses limitations found in existing single-owner models, including the absence of validation gates, inadequate governance controls, and vulnerability to single points of failure. By requiring at least two of three registered authorities to approve products before distribution, the system establishes a mandatory checkpoint that prevents unverified products from entering the supply chain. Implementation on the Sepolia testnet demonstrates a 47 percent reduction in gas consumption for product registration compared to the baseline approach, while introducing role-based access control, blacklist enforcement, and duplicate-vote prevention. Comparative analysis across eight evaluation dimensions shows improvements in trust, security, governance, validation, and auditability metrics. The system provides a transparent framework for healthcare product verification with applications across pharmaceutical and medical device supply chains.

Index Terms—Blockchain, supply chain management, healthcare verification, smart contracts, multi-authority validation, threshold-based governance, Ethereum, IPFS.

I. INTRODUCTION

The healthcare supply chain encompasses the production, distribution, and delivery of pharmaceuticals and medical devices to end consumers. Counterfeit medicines account for an estimated 10 percent of products in low- and middle-income countries according to the World Health Organization, creating a direct threat to patient safety and public health. The complexity of multi-party supply networks, combined with limited traceability across organizational boundaries, creates opportunities for product diversion, contamination, and falsification.

Blockchain technology has been explored as a potential solution to these challenges by providing an immutable, transparent ledger for tracking product provenance. The decen-

tralized nature of distributed ledgers removes reliance on a single centralized authority, while cryptographic verification ensures the integrity of recorded transactions. Within healthcare applications, blockchain systems have been proposed for tracking medicine batches from manufacturer to patient, verifying supplier credentials, and enabling real-time audit capabilities.

The work by Latif et al. introduced a blockchain-based medicine supply chain system for transparent healthcare tracking, implementing a complete lifecycle model that covers creation, sale, shipment, receipt, and consumption of medicine products [1]. This baseline system demonstrated that smart contracts can encode supply chain workflows with automatic enforcement of business rules. However, the model relies on a single-owner authorization scheme for each product, with no mechanism for independent verification or multi-party validation. Any registered actor can transfer ownership to another party without external review, and the system does not include certificate verification, governance controls, or emergency intervention capabilities.

This paper proposes an enhanced system that extends the baseline architecture with a threshold-based multi-authority validation layer. The proposed system requires products to pass an ethical review by multiple independent authorities before continuing into the distribution network. This validation gate introduces a checkpoint that prevents unverified or non-compliant products from advancing, while maintaining the transparency and immutability benefits of the underlying blockchain infrastructure. The system was implemented, deployed on the Sepolia testnet, and evaluated against the baseline model across multiple dimensions including gas efficiency, security controls, governance capabilities, and audit completeness.

II. LITERATURE REVIEW

The application of blockchain technology to supply chain management has been an active area of research since the mid-2010s. Early work focused on cryptocurrency-based payment systems, but the introduction of programmable smart contracts enabled more sophisticated tracking and verification workflows within distributed networks.

The foundational work by Latif et al. presented a blockchain-based medicine supply chain model implemented as a Solidity smart contract deployed on the Ethereum network [1]. The system models a complete product lifecycle with six sequential states: creation, sale, purchase, shipment, receipt, and consumption. Each transition is controlled by the current product owner, who initiates a transfer that the next party accepts upon verification. The contract maintains an ownership chain that can be queried at any point to trace the product history from origin to endpoint. This model established that smart contracts are capable of encoding complex supply chain workflows with automatic enforcement of business rules.

Beyond the healthcare domain, several blockchain supply chain implementations have been documented. General-purpose systems have addressed food traceability, luxury goods authentication, and automotive parts tracking [2], [3]. These systems share common architectural patterns: a smart contract layer that encodes business logic, a distributed ledger that records transactions, and a user interface that enables stakeholder interaction through wallet-based authentication. The healthcare sector presents additional requirements not present in these general implementations, including regulatory compliance, certificate verification, and multi-party authorization.

The integration of off-chain storage with blockchain systems has been addressed through IPFS-based approaches [7]. Storing large documents directly on-chain is expensive due to gas costs. Certificate documents and audit evidence can be pinned to IPFS, with only the content hash recorded on-chain. This pattern enables verifiable storage without on-chain bloat, and has been adopted in systems ranging from academic credential verification to medical records management.

Threshold-based validation mechanisms draw from established distributed systems literature on Byzantine fault tolerance and multi-party computation. The concept of requiring a minimum number of independent validators before a transaction is confirmed has been applied in consensus algorithms, but its application within supply chain smart contracts for ethical validation represents a newer direction. The proposed system adapts this pattern to a healthcare context, where product approval requires agreement from multiple independent authorities rather than a single owner decision.

III. RESEARCH GAP AND MOTIVATION

Analysis of the baseline model by Latif et al. reveals several structural limitations that reduce its suitability for production healthcare environments.

A. Single-Owner Authorization

The baseline system assigns ownership of each product to a single actor at any given time. That actor has unilateral authority to transfer the product to any other registered party. There is no requirement for independent review or approval before ownership changes hands. In a healthcare context, where regulatory compliance and product authenticity are critical, this single-owner model creates a vulnerability where a compromised or malicious actor can move counterfeit or substandard products through the supply chain without detection.

B. Absence of Validation Gates

The baseline model does not include any checkpoint where product credentials can be independently verified. Once a product is created, it flows through the supply chain based solely on owner-to-owner transfers. There is no mechanism for an authority to examine off-chain evidence such as manufacturing certificates, quality test reports, or regulatory compliance documents before approving the product for distribution.

C. Limited Governance Controls

The baseline system provides basic ownership management but lacks governance features expected in production systems. There is no capability to revoke actor privileges, blacklist participants for misconduct, pause the contract in response to emergencies, or adjust authorization thresholds. These controls are necessary for maintaining operational integrity when actors leave the network, violate policies, or when security incidents require intervention.

D. No Certificate Verification

Healthcare products require documented evidence of manufacturing standards, quality testing, and regulatory compliance. The baseline model does not provide a mechanism for storing or verifying such certificates. Without certificate anchoring, there is no link between the on-chain product record and the off-chain documentation that establishes its legitimacy.

These limitations motivate the development of an enhanced system that retains the transparency and traceability of the baseline model while introducing the governance and validation controls necessary for production healthcare deployment.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed system follows a five-layer architecture that separates concerns across the user interface, application logic, wallet interaction, blockchain execution, and off-chain storage layers.

Layer 1 — User Layer. Stakeholders include the system administrator, product manufacturer, regulatory authority, distributor, retailer, and public verifier. Each stakeholder interacts with the system through role-specific workflows determined by their assigned access credentials on the blockchain.

Layer 2 — Frontend Layer. The application is built using Next.js with the App Router architecture, TypeScript for type safety, and Tailwind CSS v4 for styling. Two primary

routes serve different user needs. The homepage provides a live wallet console for role-based product operations. The verification page enables public lookup of product details and generation of QR codes for physical product labeling.

Layer 3 — Wallet Layer. MetaMask serves as the browser-based wallet, injecting an EIP-1193 provider that the frontend uses to sign transactions and read contract state. Ethers.js v6 provides the interface between the application and the Ethereum network, handling contract ABI encoding, transaction submission, and event parsing.

Layer 4 — Blockchain Layer. Two smart contracts are deployed on the Sepolia testnet. The baseline contract implements the original paper model for comparison purposes. The proposed contract implements the enhanced system with threshold validation, role-based governance, and emergency controls. Both contracts share the same network deployment, enabling direct gas and feature comparison under identical conditions.

Layer 5 — Off-Chain Storage Layer. The Filebase IPFS RPC API provides decentralized storage for certificate documents and audit evidence. When a manufacturer uploads a certificate, the Filebase endpoint returns a content identifier that is hashed using keccak256 and stored in the product record on-chain. This pattern enables verifiable certificate storage without the gas costs of on-chain document storage.

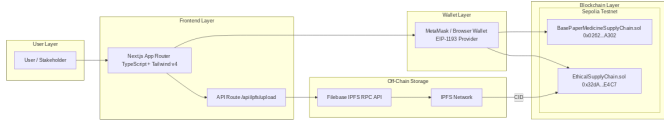


Fig. 1. System architecture showing five-layer design from user through off-chain storage.

V. THRESHOLD-BASED ALGORITHM

The core contribution of the proposed system is the Threshold-Based Multi-Authority Ethical Validation Algorithm. This algorithm introduces a mandatory checkpoint that products must pass before continuing into the distribution phase of the supply chain.

A. Algorithm Parameters

The algorithm operates with an authority threshold of two out of three, meaning that at least two independent authorities must approve a product before it is cleared for distribution. Three authorities must be registered on the contract for the system to function. The threshold value can be adjusted by the system administrator, subject to the constraint that it cannot exceed the number of currently registered authorities.

B. Algorithm Steps

The algorithm proceeds as follows. First, a manufacturer registers a product on the blockchain by providing a unique product identifier, a certificate hash derived from the IPFS-stored certificate document, and product metadata including batch number, manufacturer name, and manufacturing

and expiration dates. Second, the manufacturer advances the product from the Created state to the Manufactured state. Third, registered authorities review the off-chain certificate evidence stored on IPFS and cast votes to approve or reject the product. Fourth, after each vote, the contract checks whether the approval count or rejection count has reached the authority threshold. Fifth, if approvals meet the threshold, the product status becomes Approved. If rejections meet the threshold, the product status becomes Rejected, which is a terminal state. Sixth, the contract enforces the validation gate by blocking any attempt to advance a product from the Manufactured state unless its validation status is Approved.

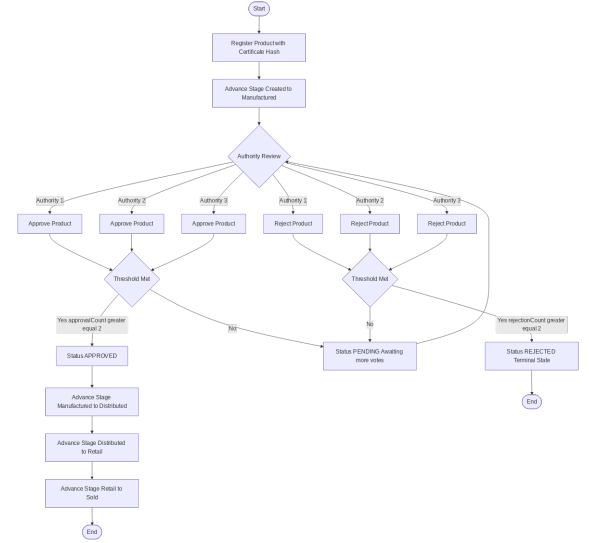


Fig. 2. Proposed system flowchart with threshold validation gate and approval/rejection paths.

C. Guard Conditions

The algorithm enforces eight guard conditions across its execution path, as summarized in Table I. Each condition prevents a specific class of unauthorized or invalid operation. The contract reverts with a descriptive error message when any guard condition fails.

D. Approval and Rejection Logic

The approval mechanism checks whether the cumulative approval count meets or exceeds the authority threshold after each authority vote. When this condition is satisfied, the product validation status transitions to Approved. The rejection mechanism operates symmetrically, transitioning the status to Rejected when the rejection count meets the threshold. The Rejected state is terminal, meaning the product cannot advance through any subsequent lifecycle stages.

E. Algorithm Pseudocode

```

function registerProduct(productId, certificateHash, meta-
data)
  require(msg.sender has MANUFACTURER_ROLE)
  require(product does not exist)
  
```

TABLE I
GUARD CONDITIONS ENFORCED BY THE VALIDATION ALGORITHM

Guard	Enforcement Point	Failure Result
Contract not paused	whenNotPaused modifier	ContractPaused revert
Actor not blacklisted	onlyActiveRole modifier	AccountBlacklisted revert
Correct role for stage	hasRole check	InvalidRoleForStage revert
Current custodian	custodian comparison	NotCurrentCustodian revert
Validation passed	status check at Manufactured	ValidationRequired revert
No duplicate votes	hasValidated mapping	AlreadyValidated revert
Validation still open	Pending status check	ValidationClosed revert

```

store Product struct with stage = Created
function approveProduct(productId)
    require(contract not paused)
    require(msg.sender has AUTHORITY_ROLE)
    require(!hasValidated[productId][msg.sender])
    hasValidated[productId][msg.sender] = true
    product.approvalCount++
    if product.approvalCount  $\geq$  authorityThreshold then
        product.validationStatus = Approved
function advanceStage(productId, nextCustodian)
    if currentStage == Manufactured then
        require(product.validationStatus == Approved)
        product.stage = currentStage + 1
        product.currentCustodian = nextCustodian

```

VI. TECHNOLOGY STACK

The proposed system was implemented using a technology stack selected for maturity, documentation quality, and compatibility with the Ethereum development ecosystem, as shown in Table II.

TABLE II
TECHNOLOGY STACK USED IN SYSTEM IMPLEMENTATION

Layer	Technology	Version
Smart Contract Language	Solidity	0.8.24
Contract Framework	Hardhat	2.24.1
Security Libraries	OpenZeppelin	v4.9.6
Frontend Framework	Next.js	16.2.1
Frontend Language	TypeScript	5
Blockchain Library	Ethers.js	6.16.0
Wallet Integration	MetaMask	EIP-1193
Off-Chain Storage	Filebase IPFS	RPC API
Target Network	Sepolia Testnet	11155111

The OpenZeppelin Contracts library provides the AccessControlEnumerable module for role-based access control, the Pausable module for emergency contract suspension, and the ReentrancyGuard module for protection against reentrancy

attacks. These components form the security foundation of the proposed contract.

VII. IMPLEMENTATION AND RESULTS

The proposed system was fully implemented and deployed on the Sepolia testnet. Both the baseline and proposed contracts operate on the same network, enabling direct comparison under identical conditions.

A. Working Features

The implementation includes twenty distinct working features. The baseline contract implements the full six-state lifecycle from the original paper. The proposed contract implements the enhanced lifecycle with threshold validation, role-based governance, blacklist enforcement, emergency pause, duplicate-vote prevention, custodian validation, next-custodian role checking, manufacturing and expiration date tracking, QR code generation for public verification, zero-address validation guards, reentrancy protection, and compact packed-struct storage for gas efficiency.

B. Deployment Details

The baseline contract is deployed at address 0x0262C4dEc9A16A5962e862926DD1AdA94c64A302. The proposed contract is deployed at address 0x32dA54F4c606fccB17fcB3f41416529b181cE4C7. Both addresses are on the Sepolia testnet with chain identifier 11155111.

C. Gas Consumption Analysis

Gas consumption was measured for three representative operations under equivalent conditions. The proposed system uses compact data types including uint8 for state enums and uint32 for timestamp fields, compared to the baseline use of strings and uint256 fields. This structural optimization produces measurable gas savings.

TABLE III
GAS CONSUMPTION COMPARISON ACROSS THREE OPERATIONS

Operation	Baseline	Proposed	Savings
Create / Register	145,056	76,085	47.6%
Sell / Advance Stage	76,558	38,450	49.8%
Buy / Approve	81,291	59,786	26.5%

The proposed system achieves a 47.6 percent reduction in gas for product registration, a 49.8 percent reduction for stage advancement, and a 26.5 percent reduction for the approval operation. The smaller savings on approval reflect the additional computational work required for threshold checking and vote recording, which the baseline does not perform.

VIII. COMPARATIVE ANALYSIS

The proposed system was evaluated against the baseline model across eight dimensions using a ten-point scoring scale. Each dimension was assessed based on the presence and quality of specific system capabilities.

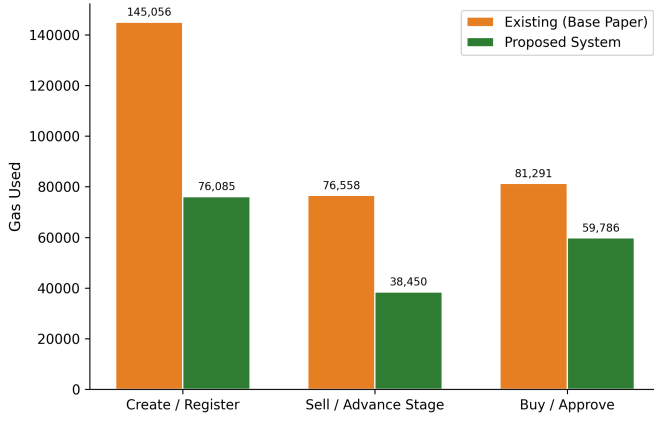


Fig. 3. Gas usage comparison across three operations between baseline and proposed systems.

A. Multi-Dimensional Evaluation

The trust dimension measures the degree to which the system prevents unilateral decision-making. The baseline scores 4 out of 10 due to its single-owner implicit trust model, where any registered owner can transfer products without oversight. The proposed system scores 9, reflecting the multi-authority explicit threshold that removes single points of failure.

Security encompasses access controls, input validation, and attack surface reduction. The baseline provides basic ownership checks and achieves a score of 5. The proposed system adds role-based access control, blacklist enforcement, contract pause capability, reentrancy protection, and zero-address validation, resulting in a score of 9.

Governance addresses the administrative controls available to system operators. The baseline provides a simple owner model without revocation safety or emergency controls, scoring 3. The proposed system includes an admin-controlled registry, authority threshold management with revocation guards, and emergency pause functionality, scoring 9.

Transparency measures the visibility of product history and validation decisions. Both systems maintain on-chain traceability, but the proposed system adds explicit validation status tracking, certificate hash anchoring, and an approval audit trail, improving from 7 to 9.

Validation represents the most significant differentiator. The baseline has no validation gate and products flow directly through the supply chain, scoring 2. The proposed system mandates threshold-based approval by independent authorities before distribution, scoring 9.

Storage efficiency evaluates the compactness of on-chain data structures. The baseline uses strings and uint256 fields throughout, scoring 5. The proposed system employs packed structs with uint8 enums, uint32 timestamps, and compact bytes32 identifiers, scoring 8.

Gas efficiency measures the computational cost of key operations. The baseline create operation costs 145,056 gas compared to 76,085 for the proposed register operation. However, the proposed system adds validation operations that

the baseline does not have, resulting in scores of 5 and 7 respectively.

Auditability addresses the completeness of on-chain records for external review. The baseline provides basic event emission, scoring 6. The proposed system emits detailed events for registration, stage advancement, approval votes, rejection votes, blacklist changes, and threshold updates, scoring 9.

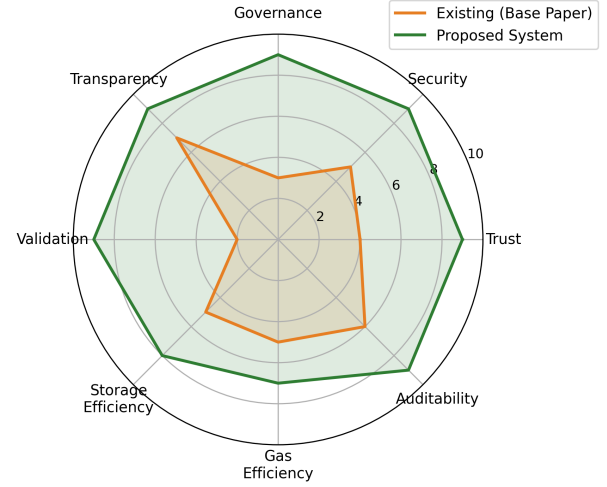


Fig. 4. Eight-dimensional comparison between existing baseline and proposed systems (scale 0–10).

B. Feature Comparison

A feature-by-feature comparison confirms that the proposed system introduces thirteen capabilities absent from the baseline model, as shown in Table IV.

TABLE IV
FEATURE COMPARISON BETWEEN BASELINE AND PROPOSED SYSTEMS

Feature	Baseline	Proposed
Multi-authority validation	No	Yes
Certificate verification	No	Yes
Blacklist enforcement	No	Yes
Emergency pause	No	Yes
Role revocation safety	No	Yes
Duplicate vote prevention	No	Yes
Custodian validation	No	Yes
Next-custodian role check	No	Yes
MFG/EXP date tracking	No	Yes
Public QR verification	No	Yes
Zero-address guards	No	Yes
Reentrancy protection	Partial	Yes
Compact storage	No	Yes

IX. CONCLUSION AND FUTURE WORK

This paper presented an enhanced blockchain-based supply chain tracking system that introduces a threshold-based multi-authority ethical validation mechanism for the healthcare sector. The proposed system addresses structural limitations

identified in the baseline single-owner model, including the absence of validation gates, inadequate governance controls, and lack of certificate verification. Implementation on the Sepolia testnet demonstrated that the enhanced system provides comprehensive security and governance capabilities while achieving significant gas savings through compact data structure optimization.

The threshold-based validation algorithm requires at least two of three registered authorities to approve products before distribution, establishing a mandatory checkpoint that prevents unverified products from entering the supply chain. The system enforces role-based access control at every lifecycle stage, prevents duplicate authority votes, validates custodian transitions, and provides emergency intervention capabilities through blacklist enforcement and contract pausing. Gas comparison results showed a 47.6 percent reduction in registration costs and a 49.8 percent reduction in stage advancement costs compared to the baseline model.

Several directions remain for future development. Camera-based QR code scanning would enable field verification of physical products without manual seed entry. A comprehensive operator dashboard with role-specific workflows would streamline the product management process for administrators, manufacturers, authorities, distributors, and retailers. Deployment to a live cloud hosting environment would enable public access beyond local development. Etherscan contract verification would improve transparency and enable third-party audit of the deployed contract source code. Extension to support additional regulatory frameworks and cross-chain interoperability would broaden the system's applicability beyond the current Ethereum-based implementation.

The proposed system demonstrates that blockchain-based supply chain tracking for healthcare products can be enhanced with multi-authority governance and threshold-based validation without sacrificing the gas efficiency and transparency that make blockchain technology attractive for this application domain.

ACKNOWLEDGMENT

The authors acknowledge the Department of Networking and Communications at SRM Institute of Science and Technology, Kattankulathur, for providing the research infrastructure and computational resources used in this work.

REFERENCES

- [1] S. Adarsh, S. G. Joseph, F. John, M. Lekshmi, and S. Asharaf, "A transparent and traceable coverage analysis model for vaccine supply-chain using blockchain technology," *IT Prof.*, vol. 23, no. 4, pp. 28–35, 2021.
- [2] U. Agarwal, V. Rishiwal, S. Tanwar, R. Chaudhary, G. Sharma, P. N. Bokoro, and R. Sharma, "Blockchain technology for secure supply chain management: A comprehensive review," *IEEE Access*, vol. 10, pp. 85 493–85 517, 2022.
- [3] A. Ahmad, N. Verma, N. Sarhan, E. M. Awwad, A. Arora, and V. O. Nyangaresi, "An IoT and blockchain-based secure and transparent supply chain management framework in smart cities using optimal queue model," *IEEE Access*, vol. 12, pp. 51 752–51 771, 2024.
- [4] G. Al-Sumaidae, R. Alkhudary, Z. Zilic, and A. Swidan, "Performance analysis of a private blockchain network built on Hyperledger Fabric for healthcare," *Inf. Process. Manage.*, vol. 60, no. 2, pp. 103 160, 2023.
- [5] P. Centobelli, R. Cerchione, P. Del Vecchio, E. Oropallo, and G. Secundo, "Blockchain technology for bridging trust, traceability and transparency in circular supply chain," *Inf. Manage.*, vol. 59, no. 7, pp. 103 508, 2022.
- [6] V. K. Chattu, "A review of artificial intelligence, big data, and blockchain technology applications in medicine and global health," *Big Data Cogn. Comput.*, vol. 5, no. 3, pp. 41, 2021.
- [7] F. Chiacchio, D. D'Urso, L. M. Oliveri, A. Spitaleri, C. Spampinato, and D. Giordano, "A non-fungible token solution for the track and trace of pharmaceutical supply chain," *Appl. Sci.*, vol. 12, no. 8, pp. 4019, 2022.
- [8] A. Ghadge, M. Bourlakis, S. Kamble, and S. Seuring, "Blockchain implementation in pharmaceutical supply chains: A review and conceptual framework," *Int. J. Prod. Res.*, vol. 61, no. 19, pp. 6633–6651, 2023.
- [9] K. Govindan, A. K. Nasr, M. Saeed Heidary, S. Nosrati-Abarghoee, and H. Mina, "Prioritizing adoption barriers of platforms based on blockchain technology from balanced scorecard perspectives in healthcare industry: A structural approach," *Int. J. Prod. Res.*, vol. 61, no. 11, pp. 3512–3526, 2023.
- [10] I. A. Omar, R. Jayaraman, M. S. Debe, K. Salah, I. Yaqoob, and M. Omar, "Automating procurement contracts in the healthcare supply chain using blockchain smart contracts," *IEEE Access*, vol. 9, pp. 37 397–37 409, 2021.
- [11] S. K. Panda and S. C. Satapathy, "Drug traceability and transparency in medical supply chain using blockchain for easing the process and creating trust between stakeholders and consumers," *Pers. Ubiquitous Comput.*, pp. 1–17, 2021.
- [12] Y. Sabri, S. Harchi, and N. El Kamoun, "Managing health supply chain using blockchain technology: State of art challenges and solution," *Int. J. Reconfig. Embedded Syst.*, vol. 11, no. 3, pp. 258, 2022.
- [13] L. Tseng, X. Yao, S. Otoum, M. Aloqaily, and Y. Jararweh, "Blockchain-based database in an IoT environment: Challenges, opportunities, and analysis," *Cluster Comput.*, vol. 23, no. 3, pp. 1–15, 2020.
- [14] Y. Wang, J. H. Han, and P. Beynon-Davies, "Understanding blockchain technology for future supply chains: A systematic literature review and research agenda," *Supply Chain Manage.*, vol. 24, no. 1, pp. 62–84, 2019.
- [15] I. Yaqoob, K. Salah, R. Jayaraman, and Y. Al-Hammadi, "Blockchain for healthcare data management: Opportunities, challenges, and future recommendations," *Neural Comput. Appl.*, vol. 34, no. 14, pp. 1–16, 2022.